Jonathan Marshall

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LinkedIn | GitHub | Portfolio | Technical Blogs

SOFTWARE DEVELOPER | FULL STACK & AI

Software developer building full-stack applications with React, Flask, PostgreSQL, and the OpenAI API. Built Ask-Flask with streaming responses, persistent sessions, session-pinned memory, and a RAG prototype using embeddings, FAISS, and PII redaction. Delivered a paid client MVP with Stripe Checkout, verified webhooks, and idempotent fulfillment. Previously managed nuclear inspection projects and cross-functional teams of more than 40 engineers and technicians.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), SQL, HTML, CSS
Frontend: React, Redux Toolkit, Vite, React Router, react-markdown, remark-gfm, PrismJS
Backend & APIs: Flask, REST APIs, SQLAlchemy (ORM/Core), Pydantic v2, Gunicorn
AI/LLM & RAG: OpenAI API (Chat Completions, embeddings), Server-Sent Events (SSE), prompt design, context management, session-pinned memory, chunking, PII redaction, FAISS, retrieval (MMR-style), simple evals/agent endpoints
Security/Observability: JWT (HTTP-only cookies), CSRF, CORS, security headers (CSP, HSTS, XFO, Referrer Policy), Flask-Limiter, structured JSON logs, request IDs, latency metrics, uniform JSON error envelopes
Data/Cloud: PostgreSQL, SQLite, AWS S3 (presigned URLs), Render
Testing/CI: Pytest, Vitest (React Testing Library), GitHub Actions, Postman

TECHNICAL PROJECTS

Ask-Flask | AI Chat App | GitHub • Live Site ***January 2026 - May 2026***

- Built a React (Vite) + Flask AI chat app integrating the OpenAI API with token streaming via SSE and a non-stream JSON fallback endpoint.
- Implemented persistent chat sessions and session-pinned memory, using summarization to maintain continuity while controlling context growth.
- Built a RAG prototype with chunking, PII redaction, embeddings, FAISS, and MMR-style retrieval. Added rate limiting, security headers, structured logs, consistent errors, and Pytest/Vitest tests with coverage thresholds.

Fantasy Football Research Hub | GitHub • Live Site ***October 2024 - July 2025***

- Built a full-stack app with React/Vite, Redux Toolkit, Flask-RESTful APIs, and PostgreSQL/SQLite via SQLAlchemy ORM.
- Implemented JWT authentication with HTTP-only cookies and CSRF protection, plus server-side AWS S3 profile uploads and signed image retrieval.

EXPERIENCE

Ecopop, Inc., Austin, TX ***January 2026 - June 2026***
Contract Software Developer | Live Product

- Built and deployed a paid client MVP on Render with React, Flask, and PostgreSQL, including a global 30-second message rotation synchronized across devices.
- Integrated Stripe Checkout with promo codes, webhook signature verification, and idempotent fulfillment to handle retries without duplicate inserts.
- Enforced a 30-message limit with deterministic FIFO eviction in the backend, including rotation and webhook fulfillment paths.

The Merrick Group, West Hazleton, PA ***February 2020 - April 2024***
Nuclear Project Manager

- Led nuclear eddy current inspection projects from requirements through team coordination and data analysis, supporting reactor safety decisions.
- Translated technical inspection findings into clear options, risks, and recommendations for nontechnical stakeholders.
- Managed concurrent projects and cross-functional teams of more than 40 engineers and technicians, coordinating schedules, constraints, and quality standards.

EDUCATION

Google AI Essentials (remote) ***August 2025 - September 2025***
Specialization (5 courses), Coursera × Google | • AI fundamentals, prompting, responsible use, productivity with AI tools
Credential: Verify on Coursera

Flatiron School (remote) ***April 2024 - September 2024***
Full-stack Web Development, JavaScript and Python Program

Virginia Wesleyan, Norfolk, VA ***September 2011 - June 2012***
Completed core Computer Science coursework